Discussion: what the evidence supports

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Discussion: what the evidence supports

The 9 studies probe distinct parts of one categorical, factor-based framework rather than repeating a single benchmark. Their common result is narrow but useful: the client construction contains its standard-Bayes limit and the server has a named project log-linear-pool recovery corner,

Discussion: what the... (part 2)

and behavior away from those limits can be measured under declared contamination, sampling, and model assumptions. The recovery identities are formal; the performance results are conditional simulation evidence.

The recovery limit is the formal anchor

The coherence of the framework rests on the KL/NLL client limits and the zero-robustness project-pool identity (eq. 10, eq. 11). This is an identity of the stated categorical implementation,

The recovery limit is the... (part 2)

not an asymptotic claim about every generalized Bayesian model or a reconstruction of the complete source protocol.

The recovery limit is the... (part 3)

The server aggregator returns the log-linear pool (eq. 9) with maximum deviation 0; the generalized posterior returns the closed-form Bayes update with deviation 5.55e-17; and the Rényi divergence, β-loss, and robust cross-entropy recover their KL/NLL limits with residuals 0, 0, and 0.

The recovery limit is the... (part 4)

Theorem 5, Corollary 6, Lemma 3, and Proposition 4 state the formal limits; the recovery checks in sec. 10.1 are the executable falsification harness.

What the study suite jointly shows

Read together, the suite separates into two kinds of result: identities that hold exactly at the standard corner, and performance contrasts that are explicitly conditional on the operating regime. The distinction is the point of the joint reading — it says which claims travel and which are tied to the declared world.

What the study suite jointly... (part 2)

Studies 1–3 sit at the standard corner. In the reduced categorical analogue, the paired difference $\Delta\bar{F} = \bar{F}_{\text{solo}} - \bar{F}_{\text{share}}$ is 3.3109 nats, so its positive sign indicates lower mean free energy with communication (sec. 10.2,

What the study suite jointly… (part 3)

fig. 12). Seeds are the independent unit; agents remain nested within seed.

What the study suite jointly... (part 4)

Dirichlet learning reduces KL from 3.4231 to 0.0027 nats (sec. 10.3, fig. 14). The configured BMR sign control favors redundant-column pruning and rejects the supported-column reduction (sec. 10.4, fig. 15).

What the study suite jointly... (part 5)

These are bounded mechanistic checks, not exact source-protocol replications or general structure-discovery results.

What the study suite jointly... (part 6)

Study 4 steps away from the corner by holding the hidden state and attack target fixed while redrawing matched contaminated colonies. The standard pool degrades across the declared rate grid.

What the study suite jointly... (part 7)

The server-side heuristic is not uniformly better: its robust members are similar to or slightly below the standard pool at low contamination, then the selected pooled display member separates in its favor under severe contamination.

What the study suite jointly… (part 8)

At the largest swept rate, AR reaches 0.9880 against 0.6928 for the standard pool (tbl. 4; the sweep plot in fig. 16 shows the separate matched-trial gallery colony, whose largest-rate separation is smaller).

What the study suite jointly... (part 9)

This regime dependence is a result, not a nuisance to be hidden: robustness can cost efficiency when the attack is weak and pay off when the declared contamination is severe.

What the study suite jointly… (part 10)

The structural extension studies (Studies 5–9) sharpen the conditional half of that split rather than adding further corner checks, and they are reported with their negative contrasts intact. Communication is not uniformly beneficial:

What the study suite jointly... (part 11)

the cross-study summary (fig. 33) reports each study's federation headline signed contrast or estimand in its native units. Its intervals come from the separate harmonized seed-level rerun, including rows whose primary figure is a deterministic diagnostic; several rows are positive,

What the study suite jointly... (part 12)

while the moving-world EFE and two-level hierarchical rows are approximately zero.

Pooling helps when views are complementary and can be unnecessary or mildly costly when the agents already agree.

What the study suite jointly... (part 13)

Adding hierarchical depth is held to the same standard — the two-level stack does not beat the flat baseline on location accuracy (the paired gap is a small, statistically reliable cost) (sec. 17.9), earning its place only by additionally resolving the context latent above chance.

What the study suite jointly... (part 14)

The joint lesson is therefore not that federation or depth is always worth its cost, but that the suite measures the regimes in which each one is.

What this simulation identifies—and what it does not

The primary robustness estimand is the matched-trial mean difference in consensus accuracy conditional on the seeded true state and attack geometry.

What this simulation... (part 2)

The 960 paired trials quantify Monte Carlo variation for that estimand; they do not average over hidden states, attack targets, adaptive adversaries, or real deployments.

What this simulation... (part 3)

The cross-study layer reduces its matched trials within seed before seed-level summaries, so clients and within-seed trials are not silently counted as independent replicates.

What this simulation... (part 4)

This follows simulation study guidance to declare the estimand and Monte Carlo unit explicitly (Morris et al. 2019; Koehler et al. 2009), and the bootstrap interval follows the declared resampling unit rather than treating nested observations as a flat sample (Loy and Korobova 2021).

What this simulation... (part 5)

The consequence is a more informative claim boundary. The sweep supports a conditional statement about this contamination mechanism and these categorical beliefs; it does not establish universal Byzantine tolerance, calibration, or truth recovery.

What this simulation... (part 6)

The result also does not identify a single universally best robustness parameter independent of the operating regime: the standard pool is preferable at the low-contamination cells in this run,

What this simulation... (part 7)

while at least one robust member has a BH-rejected positive contrast in the declared high-contamination verdict; the tied display set and deterministic tie-break are reported separately.

What this simulation... (part 8)

Those are precisely the conditions a future adaptive or deployment study must vary.

The robustness comparison is conditional and statistically qualified

The headline comparison is computed by the statistics module (sec. 10.5.2), not typed into the prose. Across 960 matched trial replicates nested within the fixed seeded world at contamination rate 0.800,

The robustness comparison... (part 2)

each robust server member is compared with the standard pool by a Wilcoxon signed-rank test (Wilcoxon 1945) and the declared family of p-values is adjusted by Benjamini–Hochberg FDR (Benjamini and Hochberg 1995).

The robustness comparison... (part 3)

A method wins only when the adjusted null is rejected and its effect is positive, here at $q = 1.11 \times 10^{-158}$ with effect size 1.0000 (large). The observed-effect power and prospective sample-size calculation are planning quantities, not independent confirmation of the result.

The robustness comparison... (part 4)

The paired design is appropriate for the controlled comparison because each condition shares the seed, true state, and attack geometry. Its interpretation remains limited: the signed-rank test concerns the distribution of paired differences, not equality of raw means,

The robustness comparison... (part 5)

and BH controls expected false-discovery proportion within the declared family rather than the probability of any false positive.

The robustness comparison... (part 6)

The confidence intervals are percentile bootstrap intervals over the matched-trial unit. Seed-level review-grid results are reported separately and do not treat the primary trial count as a population of independent worlds. These qualifications make the verdict narrower,

The robustness comparison... (part 7)

but also make it reproducible and falsifiable.

Three robustness axes remain separate

The authoritative theorem–heuristic–objective taxonomy is sec. 6.3. The results preserve it: the client theorem remains source-conditional (Mildner et al. 2025),

Three robustness axes... (part 2)

`robust_aggregate` carries only its proved recovery identity plus conditional empirical contrasts, and `variational_aggregate` carries the stated objective and raw-weight bound.

Three robustness axes... (part 3)

No figure or statistic moves a guarantee between these operators; sec. 11.13 records the complete boundary.

Three robustness axes... (part 4)

The map in fig. 9 is the compact ledger for that separation. Its fourteen rows align headline result families and public boundaries with their estimands and independent units, while its nesting strip prevents seeds, trials, agents,

Three robustness axes... (part 5)

and ordered states from being treated as interchangeable evidence. Exact estimates and intervals remain in their typed reports and the native-unit cross-study summary (fig. 33).

Accuracy and effective-weight control can be traded explicitly

The F_λ family (sec. 15.5) supplies empirical grid evidence that accuracy and effective-weight control need not be a fixed binary choice within the tested objective family.

Accuracy and... (part 2)

At $\lambda = 1.0$ the implementation recovers the current variational objective bit-for-bit;

lower explored temperatures sharpen the consensus toward the heuristic while preserving the stated raw-weight bound under its assumptions.

Accuracy and... (part 3)

This is not a derivation of the heuristic from an objective. The open problem is to identify an objective whose minimizer is both competitive across contamination regimes and accompanied by a theorem that survives beyond the present categorical construction.

Why the boundary matters downstream

The practical lesson is not that every robust rule should replace belief sharing. It is that a multi-agent active-inference system can expose separate controls for client updating, server reweighting, and objective-backed consensus,

Why the boundary matters... (part 2)

while retaining a tested route back to the ordinary pool.

Why the boundary matters... (part 3)

That separation tells a builder what can be promised: exact recovery at the named corner, conditional empirical behavior under the declared attack, and no silent transfer of a client-side or variational theorem to a different server heuristic.

Why the boundary matters... (part 4)

The result is a usable research contract for extending belief-sharing systems without confusing a simulation win with a general robustness guarantee.